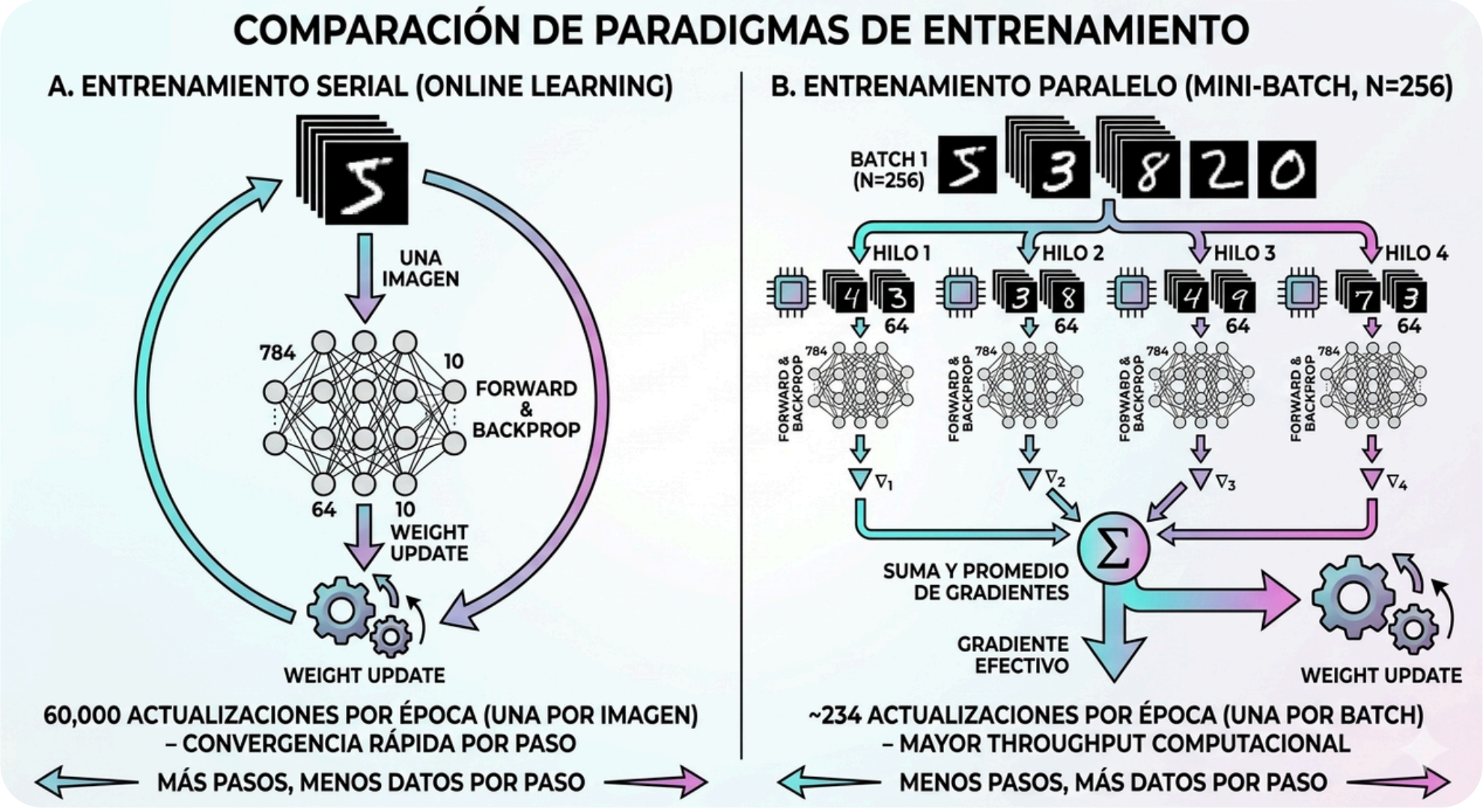
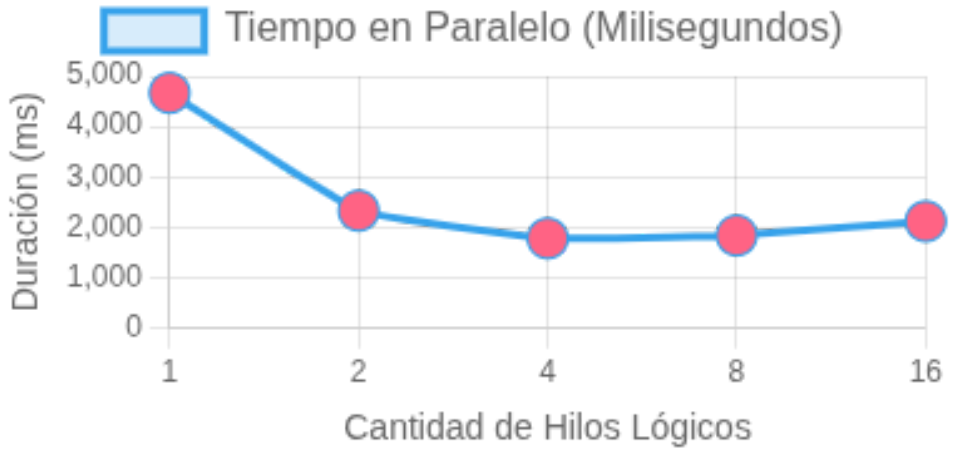
Backpropagation Paralelizado



Resultados de Tiempo

- Serial: 4437 ms
- 1 hilo: 4669 ms
- 2 hilos: 2337 ms
- 4 hilos: 1788 ms



Precisión

- Serial: 94.02%
- Paralelo: 81.66%
- Menos updates por época

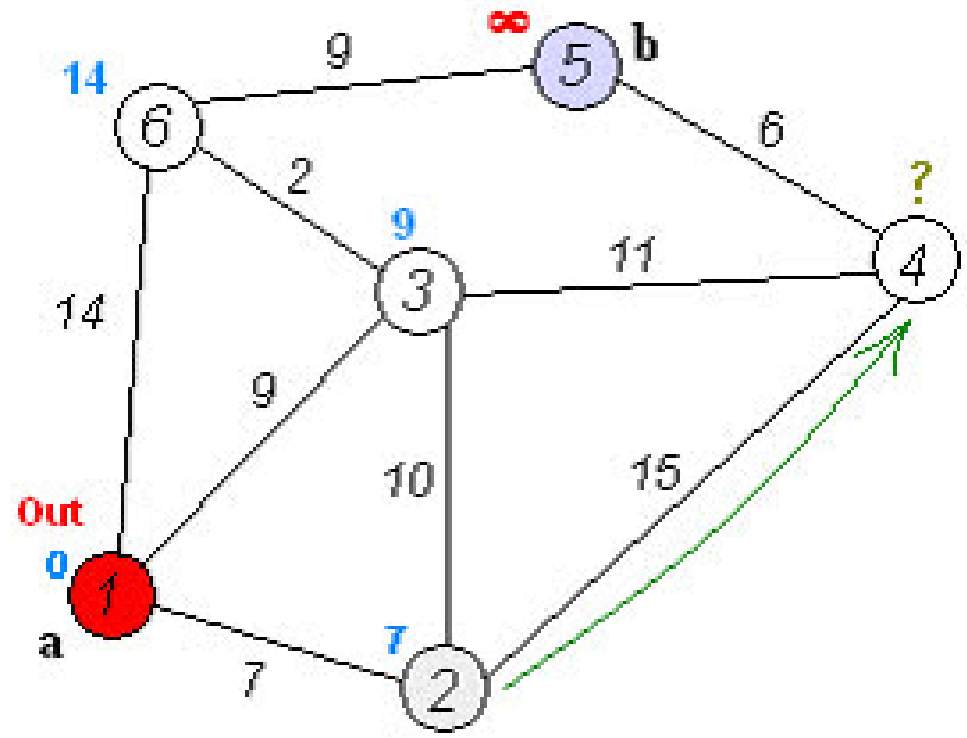
```
--- 1. ENTRENAMIENTO SERIAL (SGD) ---
Tiempo Serial: 4437 ms.
Precisión Serial: 94.02%

--- 2. ENTRENAMIENTO PARALELO (MINI-BATCH) ---
Tiempo Paralelo (1 Hilos): 4669 ms.
Precisión Paralela - 1 Hilos: 81.66%
Tiempo Paralelo (2 Hilos): 2337 ms.
Precisión Paralela - 2 Hilos: 81.66%
Tiempo Paralelo (4 Hilos): 1788 ms.
Precisión Paralela - 4 Hilos: 81.66%
Tiempo Paralelo (8 Hilos): 1846 ms.
Precisión Paralela - 8 Hilos: 81.66%
Tiempo Paralelo (16 Hilos): 2126 ms.
Precisión Paralela - 16 Hilos: 81.66%
```

Conclusiones

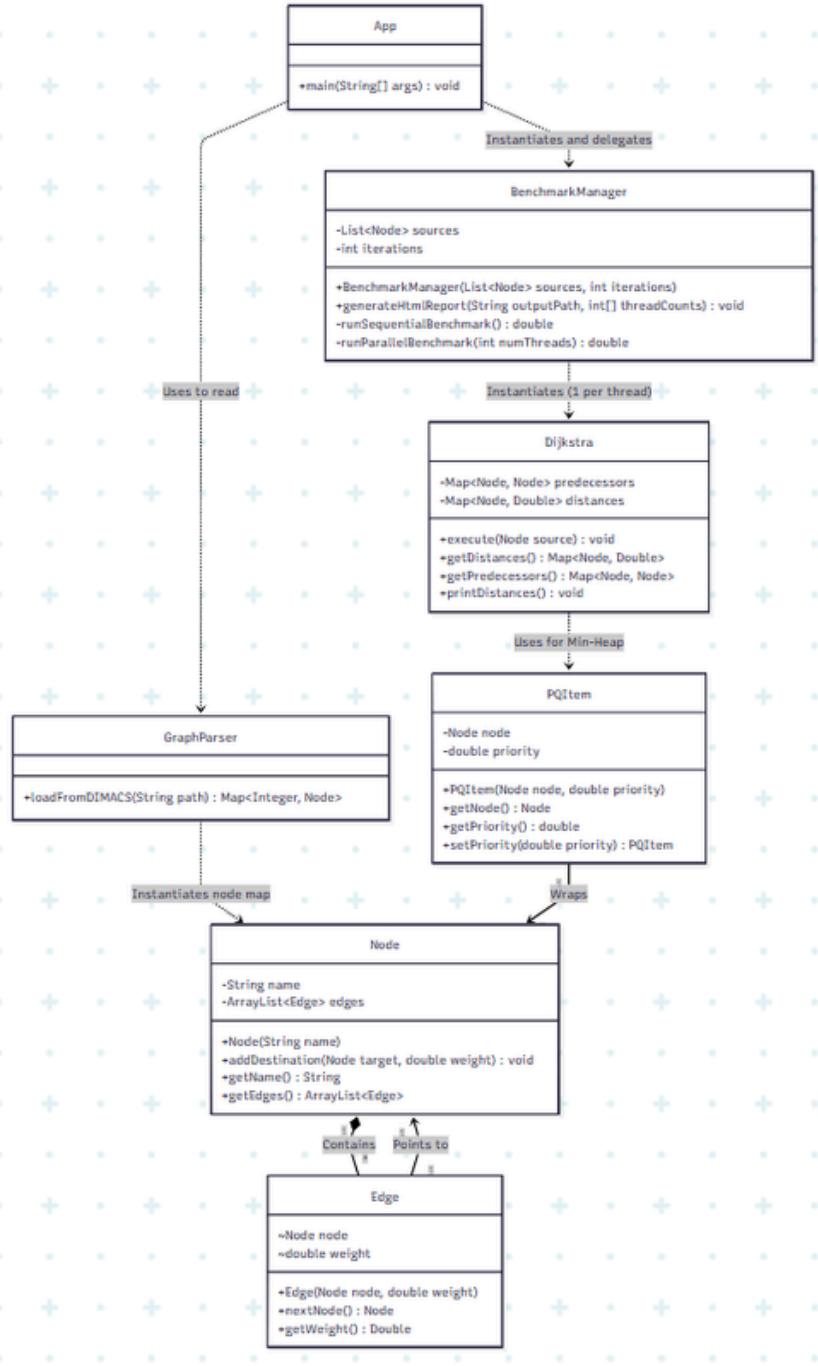
- Más velocidad (throughput)
- Trade-off clave

Parallel Dijkstra's algorithm

```
Function Dijkstra(G, s):
    dist = map initialized to ∞
    dist[s] = 0
    Q = PriorityQueue()
    Q.insert(s, 0)

    while Q is not empty:
        u, cost_u = Q.extract_min()

        if cost_u > dist[u]:
            continue

        for each (v, weight) in G.adjacent(u):
            new_cost = cost_u + weight

            if new_cost < dist[v]:
                dist[v] = new_cost
                Q.insert(v, new_cost)

    return dist
```



Architecture



Performance Report

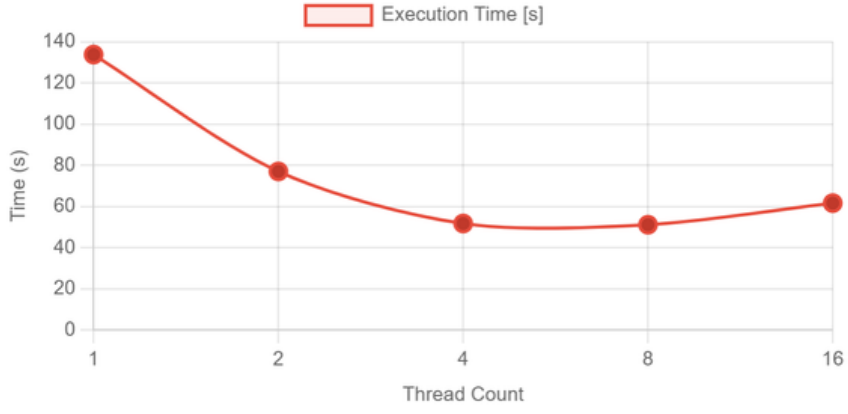
Graph Context

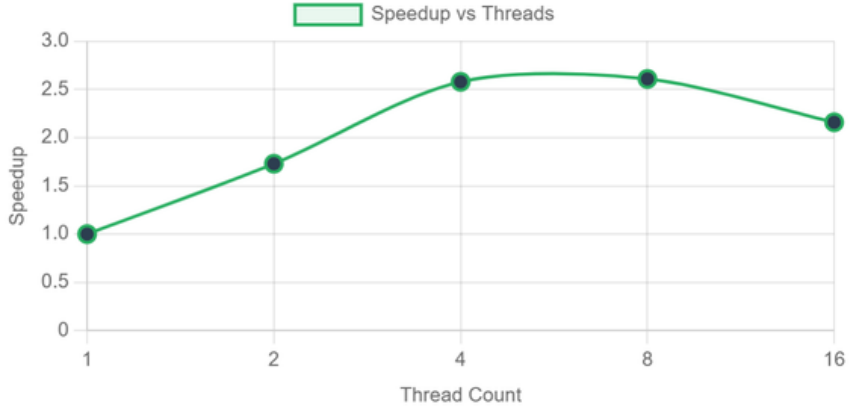
- **Total Nodes (|V|):** 264.346
- **Total Edges (|E|):** 733.846
- **Approximate Density:** 2,78 edges/node

Test Parameters

- **Evaluated Source Nodes:** 500 of 264346
- **Averaged Iterations per Test:** 3
- **Base Sequential Time (T1):** 133516,23 ms

Thread Count (p)	Parallel Time (Tp) [ms]	Speedup (S = T1/Tp)	Efficiency (E = S/p)
1	133840,56	**0,9976x**	99,76%
2	77058,45	**1,7327x**	86,63%
4	51757,94	**2,5796x**	64,49%
8	51168,69	**2,6093x**	32,62%
16	61670,69	**2,1650x**	13,53%





Conclusions

Dijkstra's algorithm parallelization using multiple sources scales efficiently only up to the limit of the processor's six physical cores.

Exceeding this threshold drastically degrades performance due to memory bus saturation and operating system context switching overhead.